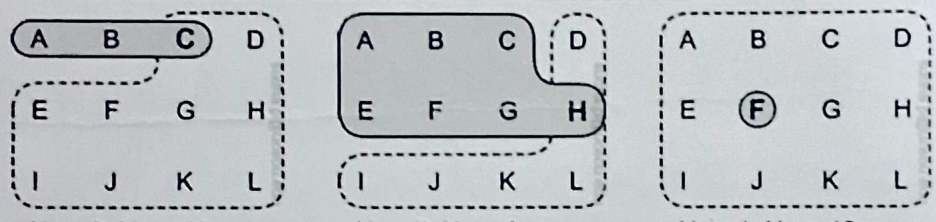
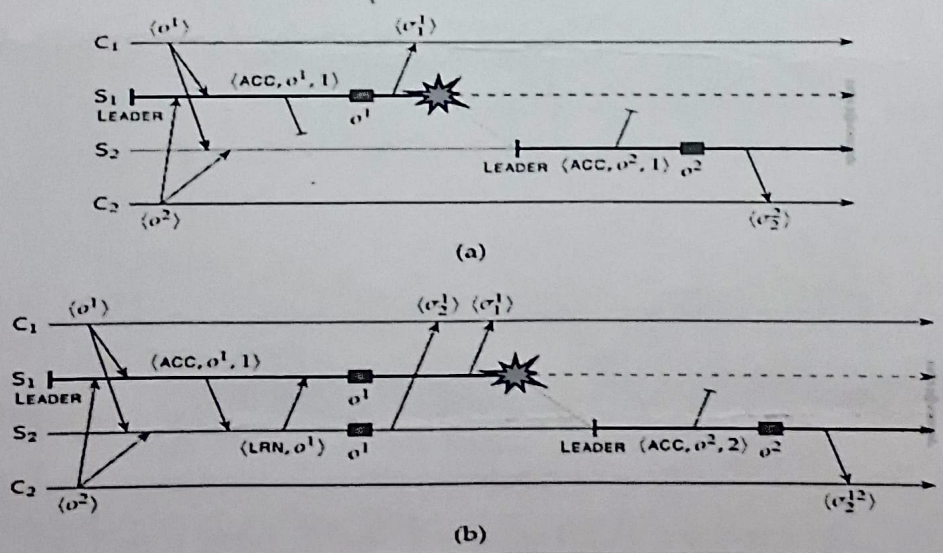
Dept . of Computer Science & Engineering

Distributed Systems – Class test –II

Time : 1.5 hour M.Marks : 30

Q.1.	Part- A (10 Marks) Answer all Questions	M	CO	BT
a)	Under what conditions does an eventually consistent system guarantee convergence? What role do update commutativity and conflict resolution play?	2	4	2
b)	How would you design a consistency model for a **real-time traffic control system**, balancing latency and correctness?	2	4	3
c)	Replica placement is a **multi-objective optimization problem**, not just a scalability decision. What are those Objectives?	2	5	3
d)	Compare **at-most-once, at-least-once,** and **exactly-once semantics** in RPC systems	2	5	3
e)	What is a **domino-effect** ? What are the Specific implications?	2	5	3
	Part- B (2 * 10= 20 Marks) Answer any TWO questions			
2. a)	Give example implementations where datastores follows – "Writes- follows Reads" consistency model and where datastores follow "Writes follow Reads" model.	5	4	4
b)	 What Problem is being solved in this figure? What are the 3 different cases shown?	5	4	4
3. a)	From the figure given below, interpret the Problem related to Failure and suggest solutions. 	6	4	3
b)	Give a brief Comparison of PAXOS Vs RAFT	4	5	3
4. a)	Analyze pessimistic vs optimistic message logging in recovery systems.	5	5	3
b)	Using finite state machine for the coordinator and for Participants in 3PC,bring out the specific advantage of using 3 Phase Commit as compared to the 2 Phase Commit Protocol for Failure Recovery.	5	5	4

Note : Give very specific and Clear answers Only.